

Power BI for Beginners

*From raw data to a live business dashboard — the restaurant way:
Extract the ingredients, prep them, cook the numbers, and plate a story.*



Extract



Transform



Load



Cook (DAX)



Serve

Why Power BI — and where we're headed

Your boss doesn't want a spreadsheet. Your boss wants answers. Power BI turns rows of raw data into pictures, KPIs, and interactive dashboards — so decisions take five seconds, not five hours.

Course roadmap: the restaurant analogy

- **Raw data** ingredients from the market — dirty, uncut
- **Power Query** the prep kitchen — wash, chop, peel
- **Data Model** the pantry — which ingredient fits which recipe
- **DAX** the cooking — new dishes from ingredients
- **Visuals & Dashboard** plating and serving the finished meal



Our one promise

You cannot break Power BI by clicking.

Click everything. Mistakes are how we learn. If something explodes, we press Undo — together.

By the final session, you will build your own full business performance dashboard — your project assignment.

Three faces, one file



Power BI Desktop

The free app on your laptop.
The kitchen — where we cook.



Power BI Service

The website in your browser.
The restaurant floor — where we share.



Power BI Mobile

The phone app.
The takeaway.

Today we live in Desktop. Recognize the neighborhoods:

Report view — the canvas — the plate

Data view — the table — the ingredients list

Model view — how tables connect — the pantry diagram

Data & Visualizations panes — your fields, and your chart types

.pbix — your whole project binder: data connection, cleaning steps, model, calculations and charts, all in one file.

Every chart must answer a question

Not decorate a page. Choose your chart with this rule of thumb:



Comparing categories

Bar or column chart



Change over time

Line chart — time moves left to right



Parts of a whole

Donut/pie — max 3–4 slices only



One hero number

Card visual — bosses love cards



Exact numbers

Table or Matrix — a pivot table in disguise

Anatomy of a visual

Axis — the categories

Values — the numbers

Legend — the color split

Drag fields into wells. The chart redraws live.

Beginner traps to avoid

- Pie charts with 10+ slices
- Rainbow colors — use 2–3 max
- Tiny, unreadable titles
- Leaving fields named "Sum of SalesAmount"

Extract, Transform, Load

Every data project on earth — a startup or a bank — runs through these three steps.



E

Extract

Going to the market and bringing raw ingredients in.

Get Data



T

Transform

The prep kitchen — wash, chop, peel. 80% of an analyst's time lives here.

Power Query



L

Load

Putting prepped ingredients into the organized pantry.

Close & Apply

Applied Steps = a video recorder. Power Query never edits your original file — it remembers your recipe and replays it. Next month's messy file arrives with the same shape? Click Refresh. All cleaning steps run alone, in one second.

How tables talk to each other

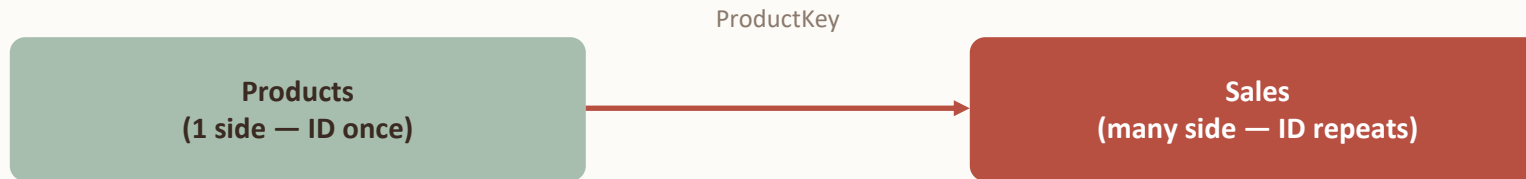
Fact tables

The events. Every row is one transaction — one sale, one order. Long, many rows, mostly numbers.
e.g. Sales

Dimension tables

The descriptions. Who, what, when. Short, wide, text-rich.
e.g. Products, Customers, Calendar

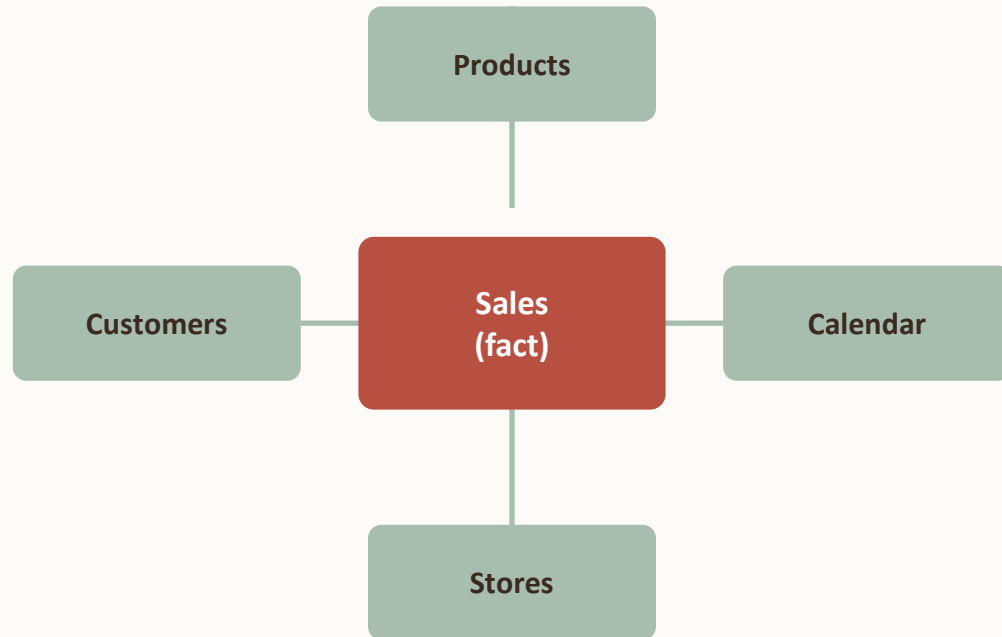
A relationship is a bridge built on a matching ID



Golden rule: filters flow from the '1' side down to the 'many' side. Click 'Bikes' in a product chart, and every number recalculates for bikes only. Keep cross-filter direction Single — 'Both' is a beginner drug that causes wrong numbers.

The star schema

One fact table in the center, dimension tables around it like rays — a star. Fast, simple, and it makes DAX easy.



Three design commandments

1**Facts in the middle, dimensions around**

Describes an event → fact. Describes a thing → dimension.

2**Always have a Calendar table**

One row per date, no gaps, no duplicates. Required for time intelligence.

3**Hide or remove unused columns**

A clean pantry cooks faster.

DAX: the language of calculated numbers

Calculated column

A new ingredient stored on every row. Computed once, stored, eats memory.

Measure (90% of your life)

A recipe computed on the fly. Not stored — recalculated for whatever the visual asks.

Filter context — the deepest idea in the course

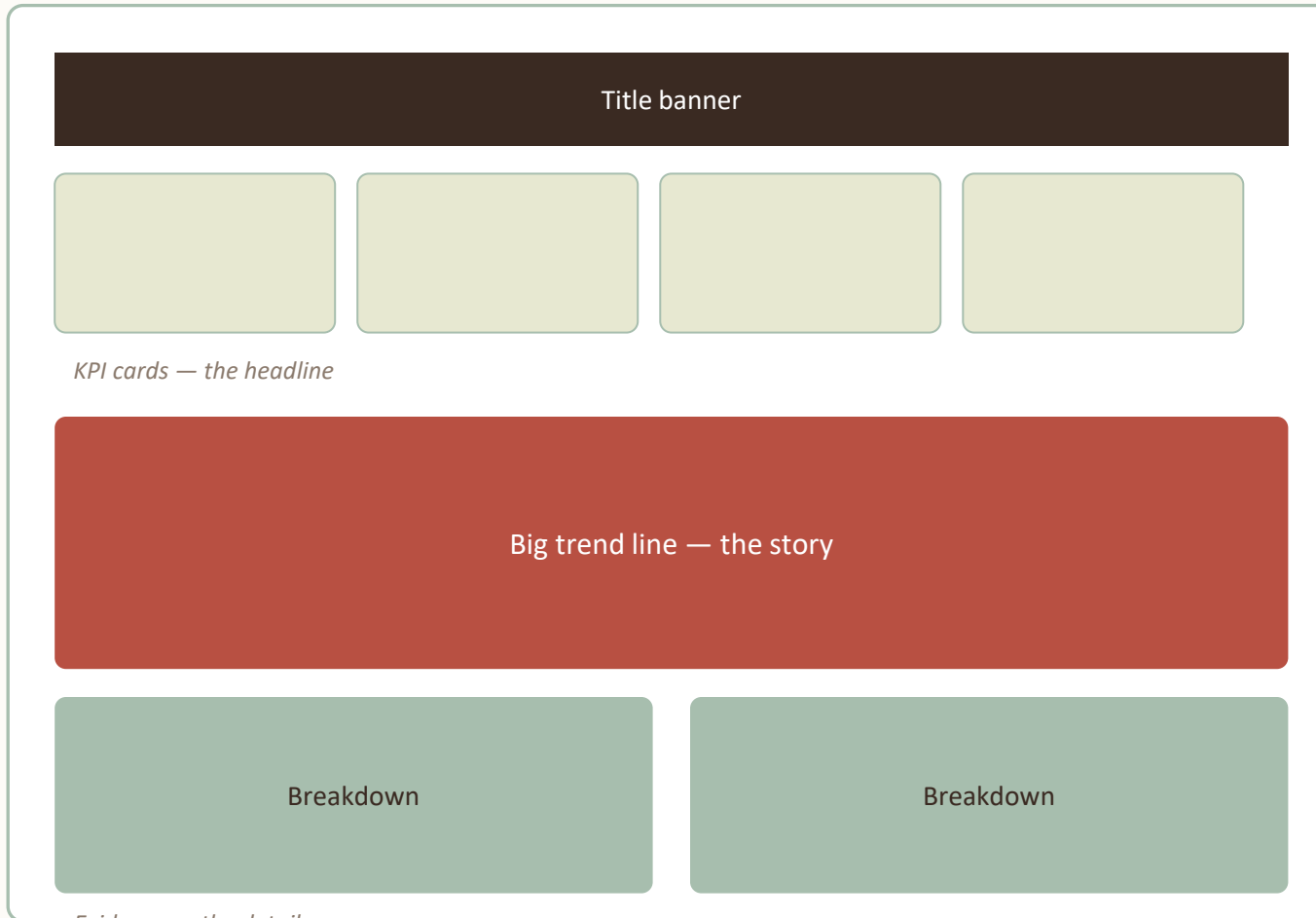
One measure, [Total Sales], has no single answer — it has an answer per question. It shows 5M in a card, 2M in the '2024' bar, 300K in the 'Bikes-West' cell. The visual asks; DAX answers.

```
Total Sales = SUM ( Sales[Amount] )  
Avg Order   = DIVIDE ( [Total Sales], COUNTROWS ( Sales ) )  
Red Sales   = CALCULATE ( [Total Sales], Products[Color] = "Red" )  
YTD Sales   = TOTALYTD ( [Total Sales], Calendar[Date] )
```

CALCULATE changes the question — every argument is a "but". **DIVIDE** is a seatbelt against dividing by zero.

Plating the meal: dashboard layout

A dashboard is a story on one screen — like a newspaper front page.



Slicer

A filter the user controls — date, region. Click it, every listening visual updates.



Filters pane

Filters you control — visual, page, or whole-report level. Business rules live here, not user choices.

2. Switch it to **Public** (so students can access without logging in) or **Private** if you want to control access.
3. Add on README file to README, delete everything from old datasets), GitHub isn't ideal — consider Git LFS or just linking to Google Drive/OneDrive from the README instead.
4. If you want to make it a bit more professional, you can use a `cd` command to copy your three folders into this directory, `git add`, `git commit -m "Add Cohort 1 class materials"`, `git push`.
5. Make a README file, copy the README file into the folder structure when you drag folders in from **Folder Explorer**.
6. Note: GitHub's web uploader has a 25MB per file limit and works best with 100 files or fewer at a time — if any folder is large, use Option B instead.
7. Replace the auto-generated README with something like this:
8. Scroll down, add a commit message like "Add Cohort 1 class materials", click **Commit changes**.
9. Option B — if you want to make it a bit more professional, you can use a `cd` command to copy your three folders into this directory, `git add`, `git commit -m "Add Cohort 1 class materials"`, `git push`.
10. Before uploading, it helps to keep the same structure you already have:

From static picture to an app

Drill-down

Click a year, fall into quarters, then months.

Tooltips

Hover and a mini-card follows your mouse.

Drill-through

Right-click a category, jump to a pre-filtered detail page.

Page navigation

Your report becomes a website with tabs.

Polish tricks that make managers say "you did this in Power BI?!"

Conditional formatting — color matrix cells green to red, like a heatmap.

Edit interactions — decide which visual is allowed to filter which other visual.



Publish sends your .pbix to the Power BI Service — share a link, schedule refresh, pin tiles online. For email: Export → PowerPoint / PDF. One cook, every serving style.

Build an interactive business performance dashboard

Cook the full meal alone. You'll receive raw, dirty, disconnected business data — nobody has cleaned or connected it yet.



Clean it

Power Query, minimum 6 recorded steps



Model it

Star schema, including a Calendar table



Calculate it

5+ measures — one CALCULATE, one time-intelligence



Present it

One page: title, KPIs, trend, 2 breakdowns, 2 slicers

Submit

Your Name_BusinessDashboard.pbix + an exported PDF or published Service link.

Final session

Three volunteers present their dashboard live — we critique kindly, like chefs tasting each other's soup.

You now have every skill of the restaurant:

Prep · Pantry · Cooking · Plating · Serving

DigiTech Academy · Data Analytics Program · Cohort 1